Lab 2 - Advanced Machine Learning

Stochastic Gradient Descent

Ludovic TRAUTMANN

CentraleSupélec

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General Informations

- **Assignment:** You will work **in pairs**.
- **Due date: Each assignment must be sent within 7 days of the lab session (by the next Thursday evening).**
  Send your notebook to `ludovic.trautmann@inria.fr` (only one submission per pair, remember to put both of your names on it).
- **Grading:** There are 6 lab sessions. The average grade will count for one third of your final grade.
  One part of the final exam will cover the material from the labs.

Lesson Recap

Motivation

Many Machine Learning problems boil down to minimizing a loss function

$$\underset{\beta}{\text{minimize}}\ \mathcal{L}(\beta) = \frac{1}{N}\sum_{i=1}^{N} l(\mathbf{x}_i, y_i; \beta)$$

Use gradient descent $\rightarrow$ What happens when N is to big ?

Stochastic Gradient Descent

Idea : We estimate the full gradient using only a part of the dataset $\mathcal{B}_t \subset [\![1, N]\!]$ (different at each iteration)

- Accurate gradient with high computational cost ?
- Fast estimation with noisy estimates of the full gradient ?

What is the optimal batchsize ?

Algorithms

Trajectory of the SGD:

$$\beta_{t+1} = \beta_t - \frac{\gamma}{\sqrt{\mathbf{s}_t + \epsilon}} m_t$$

The updates are performed coordinatewise

	Correct Position	Correct Speed
Vanilla	$\mathbf{m}_t = \mathbf{g}_t$	$\mathbf{s}_t = \mathbf{1}$
Momentum	$\tilde{\mathbf{m}}_t = \gamma \mathbf{m}_t = \theta \tilde{\mathbf{m}}_{t-1} + \gamma \mathbf{g}_t$	$\mathbf{s}_t = \mathbf{1}$
Adagrad	$\mathbf{m}_t = \mathbf{g}_t$	$\mathbf{s}_t = \mathbf{s}_{t-1} + \mathbf{g}_t^2$
RMSprop	$\mathbf{m}_t = \mathbf{g}_t$	$\mathbf{s}_t = \rho \mathbf{s}_{t-1} + (1-\rho)\mathbf{g}_t^2$
ADAM	$\mathbf{m}_t = \frac{1}{1-\theta^t}\left(\theta \mathbf{m}_{t-1} + (1-\theta)\mathbf{g}_t\right)$	$\mathbf{s}_t = \frac{1}{1-\rho^t}\left(\rho \mathbf{s}_{t-1} + (1-\rho)\mathbf{g}_t^2\right)$

The term in blue is a debiasing option